7.9 Only Five Platonic Solids

WORKSHEET

The question

STEP III 1999, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q4

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Check Euler's formula on a cube: how many vertices, edges and faces does it have? And on a square-based pyramid?

Ask yourself In a regular polyhedron, each face has m edges and each edge borders two faces. What equation does that give between E and F? Similarly, between E and V?

STAGE 2 Which values of m and n are possible?

Ask yourself What is the smallest possible value of m (sides of a face) and of n (faces at a vertex)? And since $F > 0$, what sign must h have?

Pause and try For each of the five possibilities, compute h, then F, then $E = \frac{mF}{2}$ and $V = \frac{mF}{n}$. Identify each solid.

What's really going on

Euler's formula and topology

Euler stated $V - E + F = 2$ in 1750, but his proof had gaps, and the formula has a fascinating history of attempted proofs, counterexamples and repairs. A "picture frame" polyhedron, with a square hole through the middle, has $V - E + F = 0$; a polyhedron made of two cubes joined at a vertex has $V - E + F = 3$. Imre Lakatos turned this history into a famous book, *Proofs and Refutations*, about how mathematical concepts are sharpened by counterexamples. The resolution is that $V - E + F$ depends only on the *shape* of the surface in the loose sense of topology: it is 2 for any surface that can be deformed into a sphere, 0 for a torus (a doughnut), and $2 - 2g$ for a surface with g holes. This number, the *Euler characteristic*, was one of the first ideas of topology, the study of properties unchanged by stretching and bending.

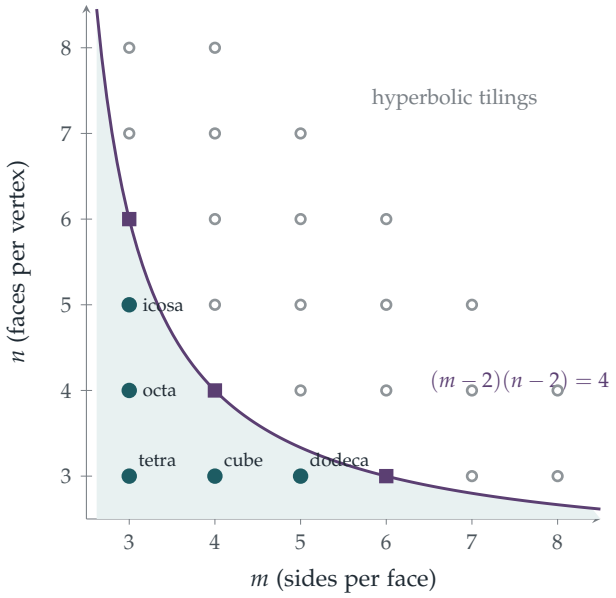


Figure 7.9.1 The possible pairs (m, n) with $m, n \geq 3$. Below the curve $(m-2)(n-2) = 4$ (shaded) are the five Platonic solids (teal). On the curve are three tilings of the flat plane (plum squares): by triangles ($m = 3$, $n = 6$), squares $(4, 4)$ and hexagons $(6, 3)$. Every point above the curve (open circles) gives a regular tiling of the hyperbolic plane.

Beyond the sphere

What happens when $h = 0$ or $h < 0$? When $(m-2)(n-2) = 4$, there are three solutions, $(3, 6)$, $(4, 4)$ and $(6, 3)$: six triangles, four squares or three hexagons meet at a point with angles adding up to exactly $360°$. These are the three regular tilings of the flat plane, the patterns of bathroom floors and honeycombs. And when $(m-2)(n-2) > 4$, the angles at a vertex add up to *more* than $360°$, which is impossible in the flat plane but perfectly possible in the hyperbolic plane of Section 5.8. There are infinitely many regular tilings of the hyperbolic plane, one for every point above the curve in Figure 7.9.1; Escher's *Circle Limit* woodcuts are drawings of some of them. So the classification of Platonic solids is really the first case of a trichotomy (sphere, plane, hyperbolic plane) that runs through all of geometry.

Symmetry groups

The symmetries of the Platonic solids form groups: 12 rotations for the tetrahedron, 24 for the cube and octahedron (which are dual: swap faces and vertices, $m \leftrightarrow n$), and 60 for the dodecahedron and icosahedron. The icosahedral group of order 60 is the "simple" group that, as we mentioned in Section 6.4, prevents the quintic from being solved by radicals. Felix Klein wrote a whole book, *Lectures on the Icosahedron* (1884), connecting the icosahedron to the solution of quintic equations.

Try this next

1 **Duality.** Show that swapping m and n swaps V and F and leaves E unchanged. Which pairs of Platonic solids are dual to each other? Which is self-dual?

2 **A football.** A traditional football is made of 12 pentagons and 20 hexagons, with three faces meeting at each vertex. Use double counting and Euler's formula to check that $V - E + F = 2$. Show more generally that any such polyhedron made of pentagons and hexagons, three at each vertex, has exactly 12 pentagons.

3 **A torus.** Make a "picture frame" polyhedron (a square block with a square hole) and count V, E and F. Check that $V - E + F = 0$.

4 **Angles at a vertex.** The interior angle of a regular m-gon is $180^\circ(1 - \frac{2}{m})$. Show that n of them fit round a vertex of a convex solid (total less than 360°) exactly when $(m-2)(n-2) < 4$, giving a second proof of the classification.

Further reading

- **David Richeson, *Euler's Gem*** (Princeton University Press). The history of $V - E + F = 2$ and how it led to topology. Excellent for a sixth-former.
- **Imre Lakatos, *Proofs and Refutations*** (Cambridge University Press). A dialogue about Euler's formula and the nature of proof.
- **H. S. M. Coxeter, *Regular Polytopes*** (Dover). The classic book on regular polyhedra and their higher-dimensional cousins.

The full guided solution is in Section 7.9:
https://beyondstep.keithmcnulty.org/ch07/platonic-solids.html